Daniel Quigley

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Statement

I work on semantics and semiotics, using mathematics and logic to determine what a change of representation preserves and what evidence licenses attributing an interpretation to a speaker or a learned system. I also study the history of such choices of representation and the philosophical consequences therein.

Postdoctoral Research

Indiana University Bloomington Bloomington, IN | Oct 2025 – Present
Postdoctoral Researcher, Center for Possible Minds

- **Principal Investigator:** Jacob Gates Foster
- Developing a vector logic that embeds typed extensional and intensional models of formal semantics homomorphically in free vector spaces. Results include a descent theorem characterizing the Boolean functionals that factor through linear maps, modal completeness theorems, exact recovery of each lower layer as a quotient or special case, and the failure of logical omniscience for belief.
- Developed a framework that audits grounding claims against a semantic standard the auditor specifies, and ran pilot audits of pretrained word2vec and of a grid-world navigator.
- Derived a longitude-resolved theory of the mean tropical year; quantified discrepancies between stated and effective seasonal targets of historical calendar reforms and constructed provably drift-bounded time-varying leap rules.
- Identified and resolved mathematical inconsistencies and consequential architecture designs in developing complex-valued BERT models for applications on quantum computers.
- Developing matched-null statistical methodology for the geometric analysis of high-dimensional embedding spaces and a spectral framework unifying principal component analysis, persistent homology, and diffusion geometry through the persistent Hodge Laplacian.
- Developing an algebra of analogy with a description-length and transfer criterion for selecting among candidate relational abstractions. Proved that the difference reading of analogical proportion is well posed exactly when the parsed observations form a torsor.
- Developing a single-sorted monoidal calculus of relations with first-order expressive power.

Education

University of Wisconsin-Milwaukee Milwaukee, WI | 2025
PhD: Linguistics

- Interdisciplinary PhD research: linguistics | mathematics | computer science | philosophy
- **Advisor:** Nicholas Fleisher (linguistics)
- **Committee:** Hamid Ouali (linguistics) | Jeb Willenbring (mathematics) | Matthew Knachel (philosophy and logic)
- **Dissertation:** *Neurosymbolic Semantics*

University of Wisconsin-Milwaukee Milwaukee, WI | 2023
MA: Linguistics

- Interdisciplinary MA research: linguistics | mathematics | computer science | philosophy
- **Advisor:** Nicholas Fleisher (linguistics)
- **Committee:** Hamid Ouali (linguistics) | Susan McRoy (computer science)

Universiteit Utrecht Utrecht, The Netherlands | 2019
MSc Certificate: Theoretical Physics

- Interdisciplinary Graduate Honors recipient
- Completed Master's coursework in Theoretical Physics and Mathematics; presented independent research on geography and climate science in the interdisciplinary honors seminar

University of Wisconsin-Madison Madison, WI | 2018
BS: Anthropology | Astronomy | Linguistics | Mathematics | Physics

- **Advisors:** Stefan Westerhoff (physics, astronomy, mathematics) | Monica Macaulay (linguistics) | J. Mark Kenoyer (anthropology)
- Completed five majors, a university record.

Research Experience

University of Wisconsin-Milwaukee Milwaukee, WI | Aug 2020 – May 2025
Graduate Researcher: Linguistics (MA 2023, PhD 2025)

- Defined a structured typology distinguishing mathematical linguistics, computational linguistics, and NLP, clarifying their scientific goals and interdisciplinary coherence.
- Developed a categorical framework that unifies formal semantic models, ensuring modular, order-independent processing.
- Proved that extensional and intensional formal semantics embed homomorphically in free vector spaces.
- Designed a generalized vector logic that realizes typed composition by multilinear maps.
- Developed a taxonomy of grounding criteria for generalized grounding problems and applied it to the symbol grounding problem in neurosymbolic AI.
- Designed algorithms for language processing based on extensional and intensional semantics and proved the underlying problems PSPACE-hard.
- Derived tensor forms for semantic representations of various linguistic phrasal types and constructions in hyperbolic spaces.

University of Wisconsin-Madison Madison, WI | Oct 2015 – May 2018
Indus Valley Civilization Undergraduate Researcher

- Reconstructed broken strings of written data from the Indus Valley Script using n-gram Markov chains and conditional entropy, and applied statistical analysis techniques to analyze the data using Python.
- Collaborated with international colleagues to analyze the Indus Valley Script data, resulting in the creation of sign frequency scores that showed the context in which different symbols were used.
- Showed that the surviving corpus supports reconstruction of broken sign strings up to length three, where earlier work had assumed longer reconstructions available.

University of Wisconsin-Madison Madison, WI | Oct 2014 – May 2018
Wisconsin Baldwin Idea Grant Project Assistant

- Coordinated with team of students and academic advisors to work with Menominee elders with Menominee language data elicitation and transcription.
- Recorded and documented language data for preservation and revitalization, and prepared teaching materials for language preservation and revitalization efforts.
- Results of the work went into a Menominee reference grammar (in development), and into a supplementary resource of glossed stories (in press).

Wisconsin IceCube Particle Astrophysics Center Madison, WI | Oct 2014 – May 2018
High Energy Astroparticle Physics Research Assistant

- Designed and implemented simulations, data acquisition systems, and visualizations for HAWC (High-Altitude Water Cherenkov) gamma-ray detector.
- Resolved discrepancies in gamma-ray results across four international experiments; wrote GPS data system in C++.
- Communicated results of simulations and technical developments with international teams.

Fellowships

NAM (Natural and Artificial Minds) Princeton, NJ | Mar 2026
Diverse Intelligences Fellow

DISI (Diverse Intelligences Summer Institute) St. Andrews, Scotland | Jul 2025
Fellow

Institute for Pure and Applied Mathematics (IPAM) Los Angeles, CA | Sep 2024 – Dec 2024
Mathematics of Intelligences Visiting Researcher

Internship Experience

Fujitsu Limited Sendai, Japan | Jun 2024 – Aug 2024
Visiting Scientist

- Collaborated on an 8-week project with Graduate-level Research in Industrial Projects for Students (G-RIPS), UCLA Institute for Pure and Applied Mathematics (IPAM) and Fujitsu Limited, focusing on explainable and interpretable AI in causal modeling.
- Reverse-engineered underlying mathematics of Fujitsu's "Wide Learning" classification machine learning model for causal AI.
- Developed a logical framework for feature relationships in Pearlean causal models.
- Developed a classifier model for causal graph structures using graph-theoretic hierarchical metrics.
- Designed an interactive, accessible interface for causal graph representation, incorporating language, vision-based, and machine-readable features for explainable and interpretable AI in causal modeling.

Apple Cupertino, CA | Jul 2023 – Nov 2023

Career Experience: Production Engineer

- Developed, tested, and deployed demo content over-the-air and operating system scripting to channel stores across 250,000 iOS, TVOS, WatchOS, and MacOS devices from development to production.
- Developed and documented sophisticated automation frameworks using Python and plist scripting to enhance operational efficiency.
- Resolved failing Wi-Fi connectivity across demo devices by investigating plist data structures for discrepancies; resolved content publishing issues by region and device.
- Validated content in twenty-one languages across twenty-five locales sensitive to local content and language requirements while crafting comprehensive test plans and technical documentation for new features and internal tools.

Conferences organized

Defining the Formal Foundations of Intelligences Bloomington, IN | May 2026

Organizing committee

- NSF-funded project called ARCANI (Accelerating Research Collaboration on Artificial and Natural Intelligences); brought together researchers from several disciplines and research networks, who share a common interest in the formal foundations of natural and artificial intelligences.
- Invited, organized, and arranged travel and accommodations for visiting conference attendees by coordinating with accounting and university hospitality.
- Arranged catering services both on and off campus.
- Presented material on holistic integration of diverse perspectives on formalisms in diverse intelligences.

Quantum AI Conference Indianapolis, IN | Aug 2026

Organizing committee

- Conference on artificial intelligence methods that use quantum and hybrid classical-quantum computing, with participants from academia, industry, health care, cybersecurity, and business.
- Organized and arranged scheduling of talks, introduction of keynote speakers, and backend logistics for travel.
- Reviewed 15 submissions for talks and poster presentations.

Presentations, Publications, Preprints

Invited talks

- **Quigley, Daniel** (2026). *A vector logic for formal semantics*. Colloquium series. Princeton University.
- **Quigley, Daniel** (2024). *Semantics as Neurosymbolic*. Linguistics/Artificial Intelligence/Mathematics. UCLA.

Presentations

- **Quigley, Daniel** (2026). *Updates since then: vector logic; analogies and abstractions; page margins*. UCLA Institute for Pure and Applied Mathematics.
- **Quigley, Daniel** (2026). *Hilbert's Apple Pie and What Does It Mean? Holistic Philosophy of Mathematics and Science, and a Haunting in Five Parts*. Formal Foundations of Intelligences. Indiana University Bloomington
- **Quigley, Daniel** (2026). *What's in a month?* Nerd Nite. Bloomington, Indiana.
- **Quigley, Daniel** (2026). *Towards analogy and abstraction via strings*. Santa Fe Institute.
- **Quigley, Daniel** (2026). *Some (coarse) geometric and topological features of semantic space: geometric invariants of word embeddings and treeish model of semantic space*. Indiana University Bloomington.
- **Quigley, Daniel** (2025). *On vector logic for extensional formal semantics*. Logic Seminar. Indiana University Bloomington.
- **Quigley, Daniel** (2025). *Symbol grounding and other grounding problems*. Indiana University Bloomington.
- Joachim, Audrey and **Quigley, Daniel** (2025). *Interdisciplinarity*. Linguistic Society of America.
- **Quigley, Daniel** and Weller, Jae (2025). *Introducing SLAC: towards inter-university community*. Linguistic Society of America.
- **Quigley, Daniel** (2024). *Semantics of imperatives and imperative constructions*. Institute for Pure and Applied Mathematics Seminar.
- **Quigley, Daniel** (2024). *Geometric page design*. UCLA Institute for Pure and Applied Mathematics Seminar.
- **Quigley, Daniel** (2024). *Syntax and Semantics of Large Language Models*. Syntax and Semantics Reading Group.
- Forde, John and Mendez, Gaspar and Okubo, Akane and **Quigley, Daniel** and Sakamoto, Renji (2024). *Fujitsu Causal Discovery: a novel interactive platform for conditional causal discovery*. Fujitsu Limited.
- **Quigley, Daniel** (2024). *Be Reasonable! Relating Logical Models and Vector Spaces for NLP Interpretability*. Workshop in General Linguistics.
- **Quigley, Daniel** (2024). *Merge: Syntax-Semantics as a Hopf Algebra*. Algebraic Structures Seminar.
- **Quigley, Daniel** (2024). *Natural Language Understanding as Tensor Product Models*. Algebraic Structures Seminar.
- **Quigley, Daniel** (2024). *A Primer on the Mathematics of Artificial Neural Networks*. Graduate Student Colloquium.

- **Quigley, Daniel** (2023). *Tensor Space and Category-Theoretic Semantics for Resolving Long-Distance Linguistic Expressions in Natural Language Processing*. PhD preliminary paper and presentation.
- **Quigley, Daniel** (2023). *Decoding Authorial Style, Tone, and Mood in Poetic Translations through Natural Language Processing: An Analysis of Beowulf*. Workshop in General Linguistics.

Workshops

- **Quigley, Daniel** (2026). *Vocabulary and mathematics of quantum mechanics for quantum computing and artificial intelligence*. University of Nottingham.
- **Quigley, Daniel** (2026). *Mathematics of shape: geometry, topology, principal component analysis, and persistent homology*. Indiana University Bloomington.
- **Quigley, Daniel** (2026). *Vocabulary and mathematics of quantum mechanics for quantum computing*. Indiana University Bloomington.
- **Quigley, Daniel** (2025). *Basics of Python (for linguists)*. UW-Milwaukee.
- **Quigley, Daniel** (2024). *On Accessibility: Screen-accessibility and PDF design*. Institute for Pure and Applied Mathematics Seminar.
- **Quigley, Daniel** (2024). *Getting Started with LaTeX*. Workshop in General Linguistics.
- **Quigley, Daniel** (2023). *LaTeX for Linguists*. Summer Workshop.

Publications

- **Quigley, Daniel** (forthcoming). *An analytic framework for historical page canons*. Journal of Mathematics and the Arts.
- **Quigley, Daniel** (2026). *page-canons: Classical page-layout canons with margin-note support*. CTAN.
- **Quigley, Daniel** (2025). *A vector logic for extensional formal semantics*. Journal of Logic, Language and Information. arXiv preprint, 2024.
- **Quigley, Daniel** (2025). *Neurosymbolic Semantics*. PhD dissertation. University of Wisconsin-Milwaukee.
- **Quigley, Daniel** et al. (2024). *White Paper: Mathematics of Intelligences*. IPAM Long Program.
- Forde, John and Mendez, Gaspar and Okubo, Akane and **Quigley, Daniel** and Sakamoto, Renji (2024). *Fujitsu Causal Discovery: a novel interactive platform for conditional causal discovery*. Fujitsu Limited.
- **Quigley, Daniel** (2024). *Be Reasonable! Relating Logical Models and Vector Spaces for NLP Interpretability*. In Proceedings: Workshop in General Linguistics.
- **Quigley, Daniel** (2023). *Decoding Authorial Style, Tone, and Mood in Poetic Translations through Natural Language Processing: An Analysis of Beowulf*. In Proceedings: Workshop in General Linguistics.

Under review

- Maynard, Eric and **Quigley, Daniel** (2026). *A framework for auditing grounding claims*. Under review: Minds and Machines. arXiv:2512.06205v3.
- **Quigley, Daniel** (2026). *A vector logic for intensional formal semantics*. Under review: Journal of Logic, Language and Information. arXiv:2602.02940v2.
- **Quigley, Daniel**. *Calendars as public temporal reference* (2026). Under review: KronoScope.
- Beckelhymer, Daniela and **Quigley, Daniel** and Flores, Cynthia and Ortega, Omayra. *From Language to Landfills: Mathematics, Ethics, and the Governance of AI*. Under review: Journal of Humanistic Mathematics.

In preparation

- **Quigley, Daniel**. *Book review: Mathematical Structure of Syntactic Merge: An Algebraic Model for Generative Linguistics by Matilde Marcolli, Noam Chomsky, and Robert C. Berwick*. In progress.
- **Quigley, Daniel**. *Determiner complexity is relative to a characterization*. Manuscript in preparation.
- **Quigley, Daniel**. *Formal framework for semiotics of writing systems*. Manuscript in preparation.
- **Quigley, Daniel**. *Mean-year arithmetic and seasonal anchors in calendar reform*. Manuscript in preparation.
- Craimer, Avi and Haydon, Nathan and **Quigley, Daniel**. *An algebra of analogies*. Manuscript in preparation.
- Craimer, Avi and Haydon, Nathan and **Quigley, Daniel**. *A relational calculus on a pairing-closed domain*. Manuscript in preparation.
- **Quigley, Daniel**. *Matched-Null Testing of Word Embedding Geometry*. Manuscript in preparation.
- **Quigley, Daniel**. *The Hodge Laplacians behind principal component analysis, persistent homology, and diffusion geometry*. Manuscript in preparation.

Preprints

- **Quigley, Daniel** (2026). *Exact semantic readout from compressed vector representations*. arXiv:2609.18047v1.
- **Quigley, Daniel** (2026). *Mean tropical year length at arbitrary ecliptic longitude*. arXiv:2605.02239v3.
- **Quigley, Daniel** (2026). *Formal specification and behavioral simulation of the holiday gift exchange game*. arXiv:2604.05219v1.
- **Quigley, Daniel** (2024). *Categorical Framework for Typed Extensional and Intensional Models in Formal Semantics*. arXiv:2408.07058v4.

Constructed languages

- **Quigley, Daniel** (2026). *Degree-type adjectives in Ahāmatya: elative, comparative, superlative, and irony*. Segments (20-02), April 2026.
- **Quigley, Daniel** (2025). *Tripartite nominalization through participant marking in Attashur*. Segments (18-01), October 2025.
- **Quigley, Daniel** (2020). *Free Genitive and Construct State in Eḥeiθymme*. FL-00006C-00, Fiat Lingua. Web. 01 September 2020.

Project Experience

Indiana University Bloomington Bloomington, IN | Oct 2025 – Present

LaTeX Development

- Reverse-engineered and derived algorithms for page layout and design according to traditional and contemporary design principles and developed LaTeX packages for modular control over text area and margin layout.
- Developed a machine-accessible custom lettrine package, conforming to WCAG level AA, for decorative initials and in-line letter decoration.
- Developed accessible graphical user interface for designing, visualizing, and implementing text area and margin layout, with LaTeX-compatible output and general typesetting parameters for use outside of LaTeX.

University of Wisconsin-Milwaukee Milwaukee, WI | Aug 2020 – May 2025

LaTeX Development

- Designed LaTeX document templates, accepted by university as official resources for graduate school.
- Created document tagging and readability methods to improve designs of accessible PDF documents.
- Developed intelligent UIs for improved accessibility of PDF documents, improving usability for users with accessibility needs and machine readability.

University of Wisconsin-Milwaukee Milwaukee, WI | May 2021

Data Visualization and Sentiment Analysis of Movie Reviews across Four Neural Network Models

- Conducted a comprehensive comparison of performance metrics for polarity sentiment analysis of movie reviews using deep learning techniques, including the design of four different machine learning architectures (CNN, RNN, RCNN, LSTM).
- Compared the performance of the different model architectures across ten epochs, with a cutoff for validation loss, and achieved an accuracy rate of greater than 83% for each model.
- Demonstrated expertise in the application of deep learning techniques to natural language processing tasks through the successful execution and analysis of this research project.

University of Wisconsin-Milwaukee Milwaukee, WI | May 2021

ML Optimization: No Free Lunch

- Optimized, evaluated, and compared performance scores for classification machine learning tasks: Decision Tree Classifier; K-Nearest Neighbor; Multinomial Naive Bayes; Logistic Regression; SVC; Dummy Classifier; Neural Network.
- Optimized, evaluated, and compared performance scores for regression machine learning tasks: Decision Tree Regressor; Linear Regression; SVR; Dummy Regressor; Neural Network.
- Evaluated CNN architectures of image classification task using the Fashion-MNIST dataset.

University of Wisconsin-Milwaukee Milwaukee, WI | Jul 2020

Linux from Scratch

- Completed *Linux from Scratch* project, building a fully functional Linux distribution from scratch using source code and following project documentation.
- Developed deep understanding of Linux operating system, including kernel, system libraries, and userland utilities, and improved skills in working with source code and building software from ground up.
- Created a customized Linux distribution that met specific needs and preferences, showcasing ability to tailor a system to meet unique requirements.

Work Experience

VersAI Milwaukee, WI | Jan 2025 – Nov 2026

Researcher

- Designed formally specified framework for LLM-based requirements verification: three-valued verdict semantics (Kleene lattice with defined collapse policies); ten evaluation primitives; ISO/IEC 25023-aligned KPIs (FRIC, VFRC, ACC, UOC) covering implementation, verification depth, and objective satisfaction.
- Developed statistical calibration layer: soundness estimation with Clopper–Pearson intervals; labeling-noise correction; importance-reweighted operational estimands subject to a minimum Kish effective sample size; per-stratum drift detection across releases.
- Authored delivery program: six-phase plan with written exit criteria; nine scoped external work packages; gold-set annotation protocol with inter-annotator agreement; governance specification for the framework's calibration hyperparameters.

Obviate Milwaukee, WI | Nov 2025 – Jun 2026

Technical Advisor; Machine Learning Scientist under intermittent contract

- Audited the operator algebra of a typed risk propagation engine for LLM-driven contract analysis; identified a sign defect on conflict edges and two further semantic errors, and derived corrected per-type operators from legal conflict typologies.
- Restored contraction without spectral normalization and designed an LLM-modulated propagation architecture with per-branch convergence guarantees.

- Delivered a 44-page technical report, an investor diligence brief, and a pre-registered validation protocol with fixed falsification conditions.

University of Wisconsin-Milwaukee Milwaukee, WI | Aug 2024 – May 2025

Research Assistant

- Designed regression models and engineered relevant features for predicting battery health in production in collaboration with Clarios, achieving 89% accuracy in predicting battery production performance.
- Developed algorithms for data explanation and interpretability of causal models and relationships for features of battery health, using LiNGAM causal models and classification.
- Developed visual representations for data, causality, and explainability for battery health and failure, presented to international team to be implemented in production.

Eruditis Milwaukee, WI | Jun 2024 – Nov 2026

Contract: Machine Learning Scientist

- Developed mathematical models for AI-enhanced algorithmic trading systems intended for non-institutional investors.
- Derived mathematical model and representation for financial metrics adaptable to investor profile parameters.
- Mentored team development of financial analyses and mathematical content.

Apple Glendale, WI | Oct 2021 – Mar 2024

Genius Technician

- Demonstrated leadership while also mentoring Technical Specialists and Technical Experts | developed and implemented new processes to improve efficiency and effectiveness of Genius Bar team.
- Exceeded expectations for customer satisfaction: attained performance review scores of 88 TMS and 74 NPS, excelling in metrics for technical expertise (89) and empathy (80).
- Certified for iPhone and Mac repair, maintaining 95% repair rate on devices.

University of Wisconsin-Milwaukee Milwaukee, WI | Aug 2021 – May 2025

Instructor of Record

- Responsible for class sizes of 20–30 students per semester, providing comprehensive support and guidance.
- Designed course content to include topics in natural language processing, such as introductory concepts and artificial intelligence ethics.
- Provided effective feedback and communication to improve performance, demonstrating commitment to student success and learning.

Graduate Teaching Experience

Linguistics 100: *Instructor of Record* Spring 2025
Natural Language Processing, Milwaukee School of Engineering: *Instructor and Advisor* Spring 2025
Linguistics 100: *Instructor of Record* Fall 2024
Linguistics 100: *Instructor of Record* Spring 2024
Linguistics 100: *Instructor of Record* Fall 2023
Linguistics 100: *Instructor of Record* Spring 2023
Linguistics 100: *Instructor of Record* Fall 2022
Linguistics 210: *Instructor of Record* Spring 2022
Linguistics 210: *Instructor of Record* Fall 2021
Linguistics 210: *Teaching Assistant* Spring 2021
Linguistics 210: *Teaching Assistant* Fall 2020

Mentoring

Brown University Providence, RI | Fall 2026

Co-mentor, Applied Mathematics

- Co-mentoring an undergraduate project in applied mathematics with Govind Menon.
- Advising on language, semiotics, and writing systems for a project that mathematically models script decipherment.

Indiana University Bloomington Bloomington, IN | Summer 2026

Co-mentor, Mathematics REU

- Co-mentored an undergraduate project in natural logic with Larry Moss.
- The student's work was presented at the department reading group and at the program's culminating conference.

Graduate Coursework

Linguistics: Phonetics | Phonology | Morphology | Syntax | Semantics | 2nd Language Acquisition | Seminar: Ellipsis | Typology and Universals | Historical and Comparative Linguistics | Seminar: Research Methods | Seminar: Double Object Constructions | Advanced Phonetics | Advanced Phonology | Advanced Syntax | Advanced Semantics | Foundations of Formal Logic | Advanced Independent Study
Computer Science: Machine Learning and Applications | Introduction to Natural Language Processing | Introduction to Artificial Intelligence | Artificial Intelligence in Business | Advanced Independent Study | Advanced Machine Learning | Algorithm Design and Analysis
Physics: Quantum Field Theory | Statistical Field Theory | General Relativity | String Theory | Field Theory in Particle Physics | Cosmology | Radiative Processes | High Energy Astrophysics
Mathematics: Differential Geometry | Geometric Partial Differential Equations | Mathematical Methods in Theoretical Physics | Algebraic Structures

Professional Interests

Linguistics: semantics | formal logic | semiotics | computational linguistics | mathematical models | model theory | ellipsis | double object constructions | transitivity mismatches | case stacking | construction grammar | scope | binding | degree and comparison | typology | language change | analogies
Computer Science: grounding problems | algorithm design | complexity | formal logic | mathematical models | model theory | machine learning methods for language processing | human language technologies | human-computer interaction
Artificial Intelligence: neural networks | natural language processing | explainable artificial intelligence | geometric neural networks | graph neural networks | semantic spaces | abstraction | analogies
Physics: gravitational physics | black hole physics | early universe physics | topological defects | quantum field theory in curved spacetime | inverse problem for Lagrangians
Mathematics: category theory | group theory | differential geometry | geometric PDEs | geometric flows | Ricci flow | operator theory | formal logic | model theory | inverse problems | abstraction
Anthropology: writing | calendrical systems | power and social relations | gender | ethnoarchaeology | archaeoastronomy
History: history of mathematics | history of logic and computing | history of linguistics | calendars | oral history

Professional Affiliations

American Mathematical Society (AMS)
American Physical Society (APS)
Association for Computational Linguistics (ACL)
Association for the Advancement of Artificial Intelligence (AAAI)
Language Creation Society (LCS)
Linguistic Society of America (LSA)
QWorld Association

Honors and Awards

University of Wisconsin-Milwaukee: *Graduate Teaching Assistantship*	Aug 2020 – May 2025
University of Wisconsin-Milwaukee: *Chancellor's Graduate Student Award*	2025
University of Wisconsin-Milwaukee: *Chancellor's Graduate Student Award*	2023
University of Wisconsin-Milwaukee: *Chancellor's Graduate Student Award*	2020
Universiteit Utrecht: *Interdisciplinary Graduate Honors*	2019

Skills

Data skills: Technical writing | Data collection, annotation, processing, visualization, statistical analysis, machine learning (Python: NumPy, Keras, Scikit-Learn, NLTK, Pandas, Matplotlib, TensorFlow, LiNGAM) | Technical documentation
Project skills: Written and oral presentation and communication | Qualitative and quantitative research methods | Problem solving | Experiment design | Language data collection and analysis | Team work and team leadership
Technical skills: Python | LaTeX | Excel | Mathematica | Praat | R | SPSS | regexp | SQL (basic) | HTML (basic) | Jekyll (basic)
Operating systems and software: Linux | Windows | MacOS | Conda | CUDA (GPU Programming) | MS Office Suite

Languages

Native: English
Elementary: German | Dutch | Finnish | Japanese
Some study: Menominee | Arabic (MSA) | Sanskrit | Georgian